1. Overall Population Trend

First dashboard contains a population bar chart from 2020-2025 and a percentage difference in population each year. Analysing the percentage difference highlights the trends or anomality's, enabling viewers to deduce any underlying social, political, or economic reasons for the changes. From 2020 to 2021 there was a -1.38% decrease in population from 2020, whereas after that population every year was increasing, albeit at a decreasing rate, from 2.16% from 2021 to 2022 to 0.58% from 2024 to 2025. The reason for the drop in population from 2020 to 2021 could be attributed to COVID 19 lockdowns. Total population of people in households in 2025 is 4,212,800, which lower than the actual population of 6.11 million (Population Brief 2025). This could be because the data does not include the homeless or Singapore citizens who are based abroad.

2. Age-Sex Breakdown

Next dashboard contains an Age Group Breakdown Bar Chart (Youth, Teen, Adult, Middle Aged Adult, Elderly), a Population Age Sex Pyramid, a Gender Ratio Line(Female to Male Ratio) and a Card showing the number of females for every 100 males. I broke down the many age groups into these 5 sections for easier analysis and to gauge age demographic trends. I also added a "Select Year" filter that toggles all the data on this dashboard other than the Gender Ratio line chart (2020-2025). Toggling between the different years, I find the general population proportion for all age groups remain the same, with highest number being Middle Aged Adults, and the lowest being Teens. Every year there is consistently more females than males, as the gender ratio line is consistently above the Equal Female to Male ratio reference line.

3. Age Group Trends

Next, I wanted to dive deeper into the Age Group Analysis, to see the changes in age demographics per year. I used a stacked area chart, and a percentage change line chart, separating the colour by age group. I made sure to use the same colours in each graph, and added an interactive legend on the right side, where users can select each age group, which will highlight the relevant parts of both charts. This will allow users to do a side by side comparison and understand the population changes each year. From looking at the percentage change line chart, I can see that the number of elderly is increasing at a faster rate compared to the other age groups. There was a 4.75% increase in elderly in 2025 while children decreased by 0.88%. This clearly shows that Singapore is experiencing an ageing population and low fertility rate.

4. Dependency Ratio

After viewing the age demographic trends, I wanted to calculate the dependency ratio, to view the economic burden on working age population(age 15-64). I created an area chart of the overall dependency ratio, allowing viewers to see that the dependency ratio has been increasing each year, from 0.46 in 2020 to 0.52 in 2025. It can be seen that the youth dependency ratio has been decreasing from 2021-2025, while from 2020-2025, every year the elderly dependency ratio increases at a rather high rate (5.05% increase from 2024-2025). This shows that the elderly ratio is increasing at such a fast rate that the overall dependency ratio is increasing, despite lower dependency from children. This shows that the ageing population is a serious economic problem in Singapore.

5. Geographical Population Distribution

This dashboard contains a map of Singapore, split into the different planning areas, with the colours representing the population level of each area, and a table that shows the number of residents in each planning area and subzone, in descending order. The map allows us to have an overview of which areas in Singapore have high population density, and which areas have lesser residents, possibly because those areas are meant for more commercial or industrial purposes. The table on the right is arranged in descending order by population level, and users can scroll down to see which areas have the lowest number of residents. From the map and the table, we can see that Bedok, Jurong West, Tampines are the top 3 most populated areas in 2020. There is also a "Select Year" filter so users can see the population distribution each year.

6. Property Type Trends

This dashboard compares the breakdown of housing types in different areas in Singapore. It consists of a bar chart comparing the percentage of government properties (HDB/ HUDC) flats, compared to private properties (Condominiums, Landed Housing, Others) in different areas in Singapore. We find that areas like Tengah, Woodlands and Jurong East have the highest percentage of government/public housing while Changi, Bukit Timah and Tanglin have the lowest percentage. A high percentage of public housing could be attributed to lower median household income in the area, and vice versa. This is because private housing is generally more expensive. It also consists of a stacked area chart split by housing which shows us that majority of Singaporeans live in Medium Size HDB Flats(3 or 4 room). The government property ratio percentage change line chart shows us that there is insignificant change in the percentage of people living in public housing yearly.